



About Me

Urban and transport planning researcher with six years of research and applied project experience. Uses spatial analysis, mobility data and policy research to examine accessibility, inequality and climate resilience in cities across Asia, Africa and North America. Specializes in age-friendly and inclusive mobility, contributing to policy reports and practical planning guidance.

Education

PhD, Urban Planning

The University of Hong Kong 2023–present
Supervisor: Prof. Jiangping Zhou. Expected February 2027.

MSc, Urban Planning

The University of Hong Kong 2019–2021
GPA: 3.59/4.3 · Student Member, Royal Town Planning Institute & Hong Kong Institute of Planners

Exchange, Urban Planning

Politecnico di Milano, Italy 2018
GPA: 30/30 · Cities, Environment and Landscape programme. China Scholarship Council award.

BEng, Urban and Rural Planning

Shenzhen University, China 2014–2019
GPA: 3.71/4.00 · Ranked 1st in the programme · Outstanding Graduate

Technical Skills

Policy & research: Policy analysis and evaluation; surveys and interviews; research reports and policy briefings.

Data & spatial analysis: Statistical modelling and machine learning (Python, R, SPSS); mapping, accessibility and network analysis (ArcGIS Pro, QGIS).

Planning & design: Urban and landscape design; drafting, 3D modelling and rendering (AutoCAD, SketchUp, Rhinoceros, Lumion).

Visual communication: Data visualization, diagrams and presentation design; report layout and image editing (Adobe Illustrator, InDesign, Photoshop).

Languages

Mandarin Chinese: Native

English & Cantonese: Full professional proficiency

Japanese & Italian: Elementary proficiency

Research & Professional Experience

• Senior Research Assistant

Station-Area Mobility and Wellbeing

The University of Hong Kong

Nov 2025–present

- **Integrated mobility analysis:** Analyze and visualize mixed-method evidence on station pairs, active-mode connections, travel patterns and rider wellbeing for an RGC General Research Fund project (17602624).
- **Planning applications:** Identify gaps in walking and cycling access around stations to inform first- and last-mile investments and station-area planning.

• Doctoral Researcher

Transit Equity and Inclusive Mobility

The University of Hong Kong

Feb 2023–present

- **Age-friendly Accessibility:** Combined travel survey data and pedestrian network analysis to assess older adults' access to everyday services, identifying gaps overlooked by a uniform 15-minute benchmark.
- **Inclusive public transport:** Analyzed large-scale metro smart-card data to identify age-related travel-time disparities, accounting for differences in travel frequency and time of travel.
- **Transit equity:** Investigate inequalities in older adults' access to transport and everyday activities, examining how accessibility and social disadvantages relate to activity participation.
- **Climate change & urban inequality:** Co-authored studies on unequal impacts of extreme heat on subway use in New York, urban heat around metro stations in Shanghai, and links between street networks and poverty in Nairobi.

• Research Assistant

Smart Mobility and Cross-Boundary Transport

HKU-Shenzhen Institute of Research and Innovation

May 2021–Nov 2023

- **Cross-boundary mobility:** Examined travel demand, travel behavior and smart mobility services across Hong Kong, Zhuhai and Macao through research reporting for the Hong Kong–Zhuhai–Macao Bridge project.
- **Research synthesis:** Prepared reports bringing together findings on cross-boundary travel and service needs for a project supported by China's National Key R&D Program (2019YFB1600703).

• Research Assistant

Intermodal Transport Data-Sharing

The University of Hong Kong

Jul 2020–Feb 2021

- **Policy benchmarking:** Benchmarked integrated transport, mobility data governance and demand-responsive transport across Copenhagen, London, Paris, Sydney, Seoul, Shenzhen and Chicago.
- **Policy reporting:** Contributed analytical briefings and report chapters to the Intermodal Transport Data-Sharing Program, funded by Hong Kong's Innovation and Technology Fund (GSP/016/19).

• Urban Planning and Design Intern

Planning & Design Projects

Research Institute of Harbin Institute of Technology

Aug–Oct 2018

- **Architectural design:** Contributed to design proposals and prepared architectural drawings and presentation materials for project reviews.
- **Local planning engagement:** Conducted field investigations and presented village renewal and master-planning proposals to county stakeholders.

Policy & Applied Projects

Integrated Urban Transport Governance

2026 | Book Manuscript Contributing author

Researched and drafted sections for two chapters of *Policy Analysis for Integrated Urban Transport Governance*, developed in collaboration with Tongji University and currently in copyediting.

Urban Mobility Handbooks

Policy & Practice Guides

2024–2025 | Contributing author

Contributed writing to two practitioner handbooks published through the 9.22 Green Mobility Platform: *Shared Public Space (2024)*, on connecting public-space planning with walking and people-centered streets; and *Inclusive Mobility (2025)*, on universal design and inclusive transport services.

City Connect 15+

Participatory GIS

2023 | Project Team Member, Concept Development & Visual Design

Co-developed the project concept and visual communication for a public-facing participatory GIS tool. City Connect 15+ integrated a 3D pedestrian network with user-defined walking speeds and weather conditions to assess access to eight categories of everyday amenities and collect mapped feedback on walking routes.

Merit Award, Hong Kong Common Spatial Data Infrastructure Awards, Open Category.

Talks & Knowledge Exchange

Conference Presentations

- Dec 2024 | Harbin Institute of Technology, Shenzhen
Invited PechaKucha presentation: *Mind the gap: From research to age-friendly design.*
- Jul 2024 | IACP Annual Conference, Hangzhou
Presenter and session chair: *From proximity to accessibility: An ageing perspective of 15-minute city planning in Hong Kong.*
- Jan 2024 | TRB Annual Meeting, Washington, DC
Oral presentation: *Toward age-friendly station design: Uncovering indoor mobility challenges for elderly metro riders.*
- May 2023 | Smart Cities Symposium Prague
Oral presentation: *Age-friendly public transport: Evidence from older adults in Hong Kong's metro.*

Invited Guest Lectures

- Jul 2024 | Shanghai University of Finance and Economics
Geospatial analysis with GIS and Python for public policy.
- Jun 2024 | Tongji University
Transport policy appraisal, multi-criteria methods and quasi-experimental designs.
- Jul 2023 | Southwestern University of Finance and Economics
Benchmarking in policy analysis and planning.

Selected Publications

12 peer-reviewed publications, 3 as first author. Five recent publications selected for policy relevance; full list: [Google Scholar](#)

- Rethinking the X-minute city for older adults: Temporal thresholds, hierarchical needs and age-specific walking accessibility

Lei, S., Zhao, M., Zhou, J., Yang, X. (2026). *Journal of Transport Geography* 134, 104656.

Most daily walking trips by older adults in Hong Kong take 5–12 minutes depending on age and destination, showing that a uniform 15-minute benchmark substantially overstates accessibility for older people.

- Evaluating age-friendly public transportation: A novel smart card data approach analyzing travel time differences between older and younger travelers after controlling for confounding effects

Lei, S., Zhou, M., Deng, W., Tang, Y., Zhou, J. (2026). *Transportation Research Part A: Policy and Practice*, 204, 104842.

Identifies metro stations and routes where older riders face the largest travel-time gaps, especially in weekday morning peaks, to prioritize age-friendly upgrades.

- Disparities in the impacts of extreme heat events on subway ridership in New York City

Zhao, M., Lei, S., Li, S., Li, W. (2026). *npj Sustainable Mobility and Transport* 3, 17.

Seniors and riders with disabilities showed the largest heat-related ridership declines; low-income Fair Fares riders had the least scope to shift travel away from the hottest hours.

- Incorporating urban thermal comfort into transit-oriented development planning: Non-linear heterogeneous built environment effects

Zhao, M., Lei, S., Li, W. (2026). *Sustainable Cities and Society* 140, 107254.

Density, a core principle of transit-oriented development, carries a thermal cost: higher population and building density around Shanghai metro stations are associated with worse summer comfort, while varied building heights and vegetation improve it.

- Association between multidimensional poverty and urban spatial network design: Comparison between theory-driven and data-driven lenses

Gachanja, J., Lei, S., Adero, N. (2025). *Applied Geography* 178, 103578.

In Nairobi, formal or informal settlement status remains the strongest contextual factor associated with multidimensional poverty, while network density, efficiency and severance add explanatory value.

Academic Service

Journal Peer Review

2026 | Verified on ORCID

Sustainable Cities and Society; Journal of Transport & Health; Journal of Transport Geography; Urban Rail Transit.